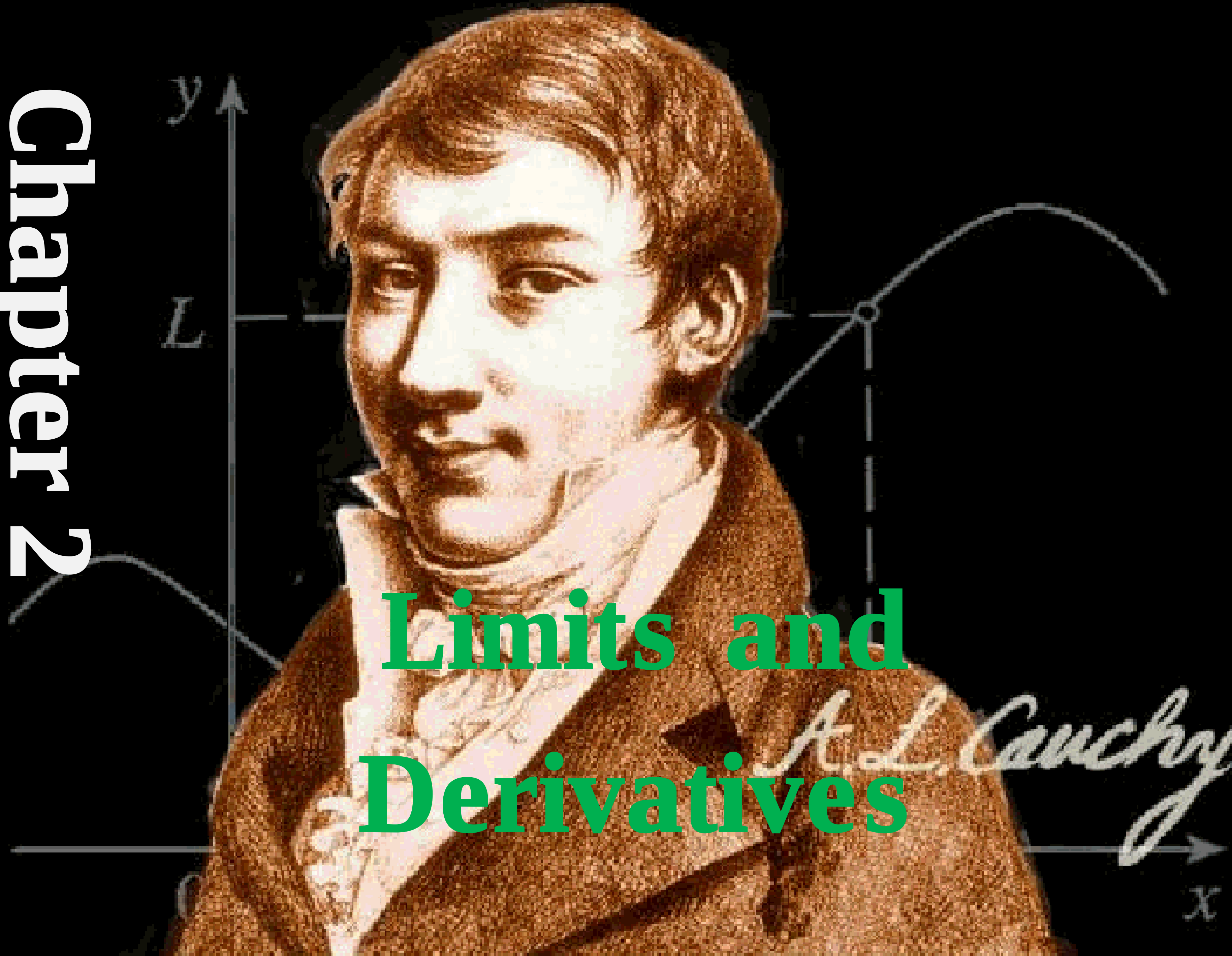
UCCM1153

Introduction to Calculus and Applications

Chapter 2

Limits and Derivatives



Section 2.1

Limits

Concept of Limit

Let $f\,(x)$ be a function and c as a constant. Suppose that $f\,(x)$ is defined at every point (except possibly at c) on the interval (r, s) containing c. If there exists a unique number L such that $f(x)$ **approaches** c, then we say that L is *the limit of* $f(x)$ *as* x *approaches* c and write as $\lim_{x \to c} f(x) = L$

An alternative notation for $\lim_{x \to c} f(x) = L$ is

$f(x) \to L$ as $x \to c$.

Concept of Limit

- For example, $f(x) = 2x+3$

Limit Numerically

x	$f(x)$
2	7
3.6	10.2
3.9	10.8
3.99	10.98
3.999	10.998
...	...
4.001	11.002
4.01	11.02
4.1	11.2
4.8	12.6
5	13

These inputs approach 4 from the left.

4

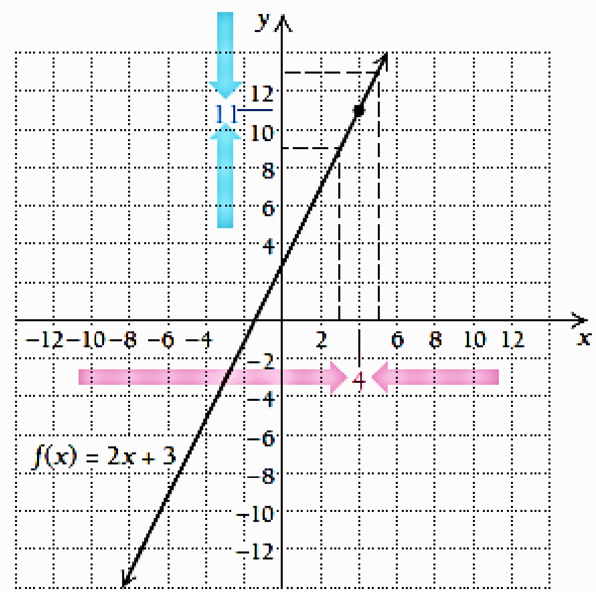
These inputs approach 4 from the right.

These outputs approach 11.

11

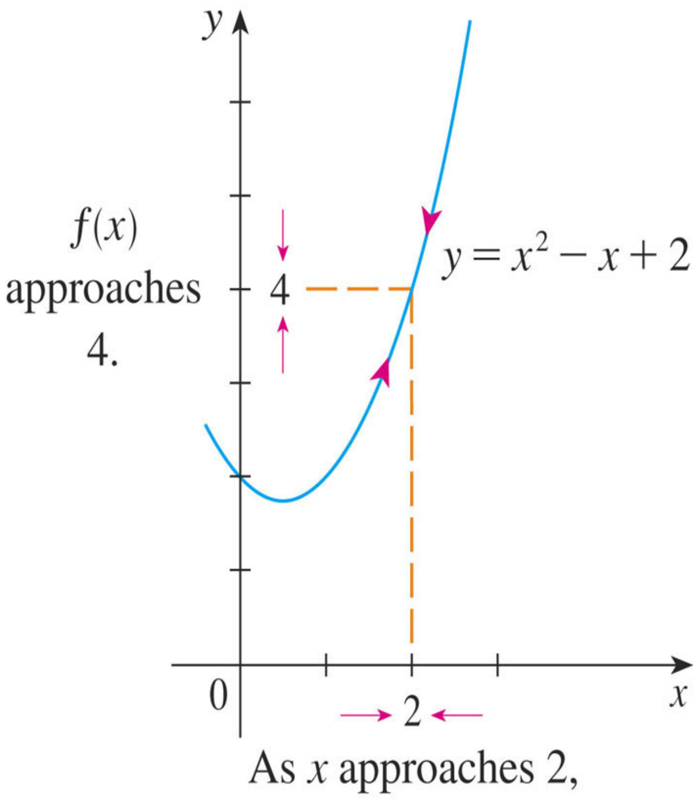
These outputs approach 11.

Limit Graphically



Concept of Limit (Cont')

Example:



Limit Graphically

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

Limit Numerically

Concept of Limit (Cont')

Example (Cont')

- Let's investigate the behavior of the function *f* defined by $f(x) = x^2 - x + 2$ for values of x near 2.
- The following table gives values of $f(x)$ for values of x close to 2, but not equal to 2.
- From the table and the graph of *f* (a parabola) shown in the figure, we see that, when x is close to 2 (on either side of 2), $f(x)$ is close to 4.
- In fact, it appears that we can make the values of $f(x)$ as close as we like to 4 by taking x sufficiently close to 2.
- We express this by saying "the limit of the function $f(x) = x^2 - x + 2$ as x approaches 2 is equal to 4."
- The notation for this is:

$$\lim_{x \to 2} (x^2 - x + 2) = 4$$

Concept of Limit (Cont')

***Example* 2.1**

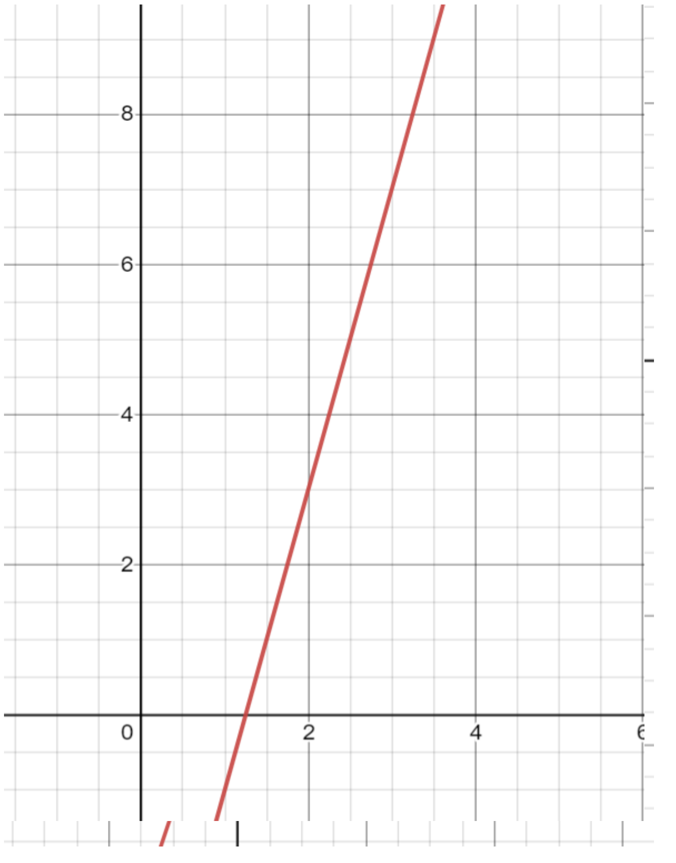
Find $\lim_{x \to 1}(3x - 4)$.

Answer: -1

***Example* 2.2**

Find $\lim_{x \to 3}(4x - 5)$.

Answer: 7



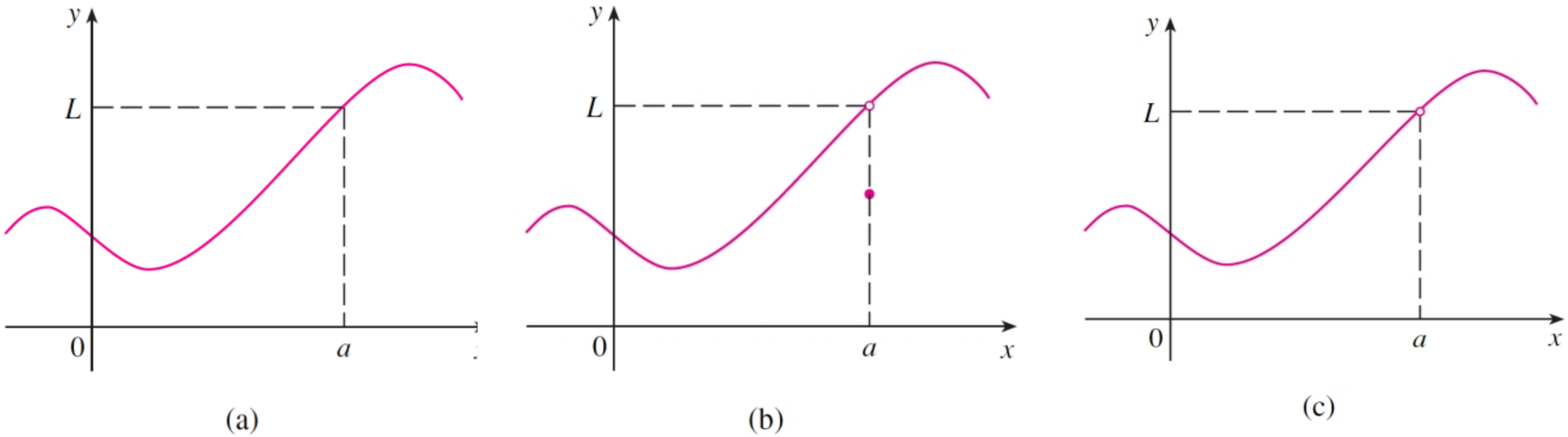
Concept of Limit (Cont')

- Three cases for $\lim_{x \to a} f(x) = L$ exists:

(a) $f(a) = L$ and $\lim_{x \to a} f(x) = L$

(b) $f(a) \neq L$ but $\lim_{x \to a} f(x) = L$

(c) $f(a)$ not defined but $\lim_{x \to a} f(x) = L$



- The above graphs of three functions. Note that…
  - in part (c), $f(a)$ is not defined;
  - in part (b), $f(a) \neq L$.
- But in each case, regardless of what happens at a , it is true that $\lim_{x \to a} f(x) = L$

Methods used to find Limits

(1) Direct substitution

***Example* 2.3**

$$\lim_{x\to 1}\frac{2x-1}{2x+1}=\frac{2(1)-1}{2(1)+1}=\frac{1}{3}$$

(2) Factorization

***Example* 2.4**

$$\lim_{x\to 1}\frac{x-1}{x^2-1}=\lim_{x\to 1}\frac{(x-1)}{(x-1)(x+1)}=\lim_{x\to 1}\frac{1}{(x+1)}=\frac{1}{1+1}=\frac{1}{2}$$

TRY: $$\lim_{x\to 3}\frac{x^2-9}{x^2-10x+21}$$

Methods used to find Limits (Cont')

(3) Multiplication with the conjugate of irrational expression

- Note: Given $\sqrt{a}+b$, its conjugate is $\sqrt{a}-b$.

Example **2.5**

Find the $\lim\limits_{x\to 0}\dfrac{\sqrt{x^2+9}-3}{x^2}$.

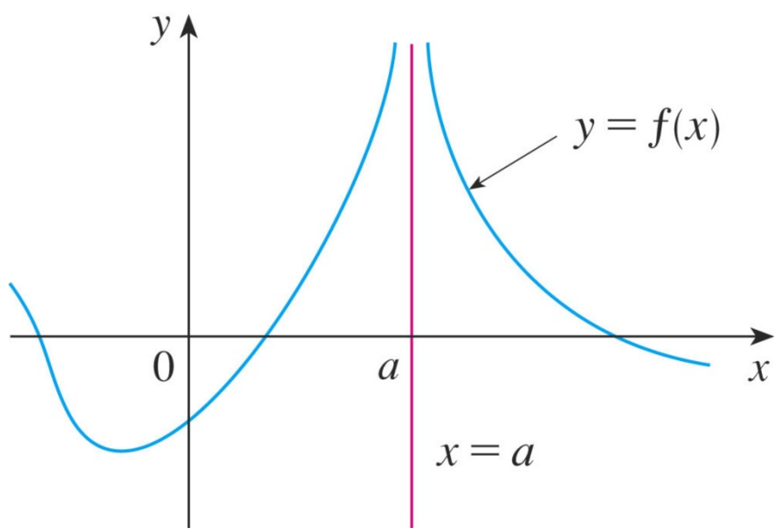
$$\lim_{x\to 0}\frac{\sqrt{x^2+9}-3}{x^2}\cdot\frac{\sqrt{x^2+9}+3}{\sqrt{x^2+9}+3}=\lim_{x\to 0}\frac{x^2+9-9}{x^2\left(\sqrt{x^2+9}+3\right)}=\lim_{x\to 0}\frac{1}{\sqrt{x^2+9}+3}=\frac{1}{6}$$

Try: $\lim\limits_{x\to 4}\dfrac{\sqrt{x+5}-3}{x-4}$

Methods used to find Limits (Cont')

(4) Graphically

For $\lim_{x \to a} f(x) = \infty$ is $f(x) \to \infty$ as $x \to a$:



One-sided Limits

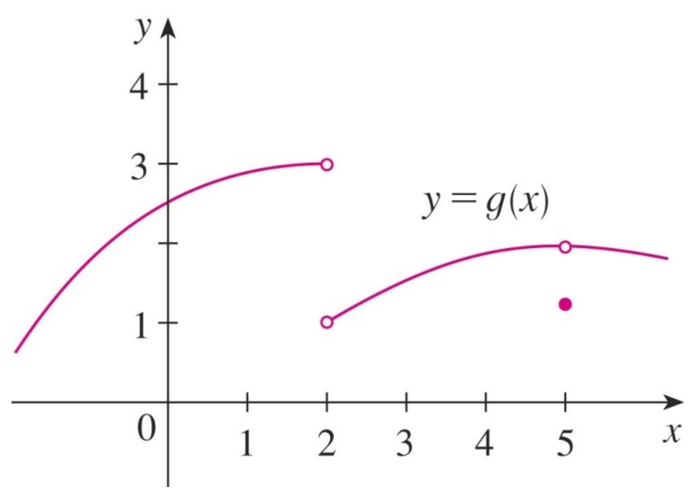
(1) Left-hand limit of *f*(*x*) as *x* approaches *c*, $\lim_{x \to c^-} f(x) = L$

(2) Right-hand limit of *f*(*x*) as *x* approaches *c*, $\lim_{x \to c^+} f(x) = L$

Note:

- $\lim_{x \to c} f(x) = L$ if and only if $\lim_{x \to c^-} f(x) = L$ and $\lim_{x \to c^+} f(x) = L$

***Example* 2.6**



One-sided Limits (Cont')

***Example* 2.6 (Cont')**

- From the graph, we see that the values of $g(x)$ approach 3 as x approaches 2 from the left, but they approach 1 as x approaches 2 from the right. Therefore, $\lim_{x\to 2^-} g(x) = 3$ and $\lim_{x\to 2^+} g(x) = 1$.

- As the left and right limits are different, we conclude that $\lim_{x\to 2} g(x)$ does not exist.

- The graph also shows that $\lim_{x\to 5^-} g(x) = 2$ and $\lim_{x\to 5^+} g(x) = 2$.

- For $\lim_{x\to 5} g(x)$, the left and right limits are the same.

- So, $\lim_{x\to 5} g(x) = 2$.

The Limit Laws

- Suppose that c is a constant and the limits $\lim_{x \to a} f(x)$ and $\lim_{x \to a} g(x)$ exist. Then

1) $\lim_{x \to a} [f(x) + g(x)] = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$
2) $\lim_{x \to a} [f(x) - g(x)] = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$
3) $\lim_{x \to a} [c f(x)] = c \lim_{x \to a} f(x)$
4) $\lim_{x \to a} [f(x) g(x)] = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$
5) $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)} \quad if \lim_{x \to a} g(x) \neq 0$

The Limit Laws (Cont')

6) $\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n$

7) $\lim_{x\to a} c = c$

8) $\lim_{x\to a} x = a$

9) $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$

10) $\lim_{x\to a} x^n = a^n$

The Limit Laws (Con't)

***Example* 2.7**

Given that $\lim_{x \to -2} f(x) = 1$ and $\lim_{x \to -2} g(x) = -1$, find

(i) $\lim_{x \to -2} [f(x) + 5g(x)]$ (ii) $\lim_{x \to -2} \dfrac{2f(x)}{5g(x) - 4}$

$$\lim_{x \to -2} [f(x) + 5g(x)]$$

$$= \lim_{x \to -2} f(x) + \lim_{x \to -2} [5g(x)]$$

$$= \lim_{x \to -2} f(x) + 5 \lim_{x \to -2} [g(x)]$$

$$= 1 + 5(-1) = -4$$

$$\lim_{x \to -2} \frac{2f(x)}{5g(x) - 4}$$

$$= \frac{\lim_{x \to -2} 2f(x)}{\lim_{x \to -2} 5g(x) - 4}$$

$$= \frac{2[1]}{5[-1] - 4}$$

$$= -\frac{2}{9}$$

The Limit Laws (Con't)

Example 2.8 (Con't)

$$\lim_{x \to 2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

$$\lim_{x \to -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$

$$= \frac{\lim_{x \to -2} (x^3 + 2x^2 - 1)}{\lim_{x \to -2} (5 - 3x)}$$

$$= \frac{\lim_{x \to -2} x^3 + 2 \lim_{x \to -2} x^2 - \lim_{x \to -2} 1}{\lim_{x \to -2} 5 - 3 \lim_{x \to -2} x}$$

$$= \frac{(-2)^3 + 2(-2)^2 - 1}{5 - 3(-2)} = -\frac{1}{11}$$

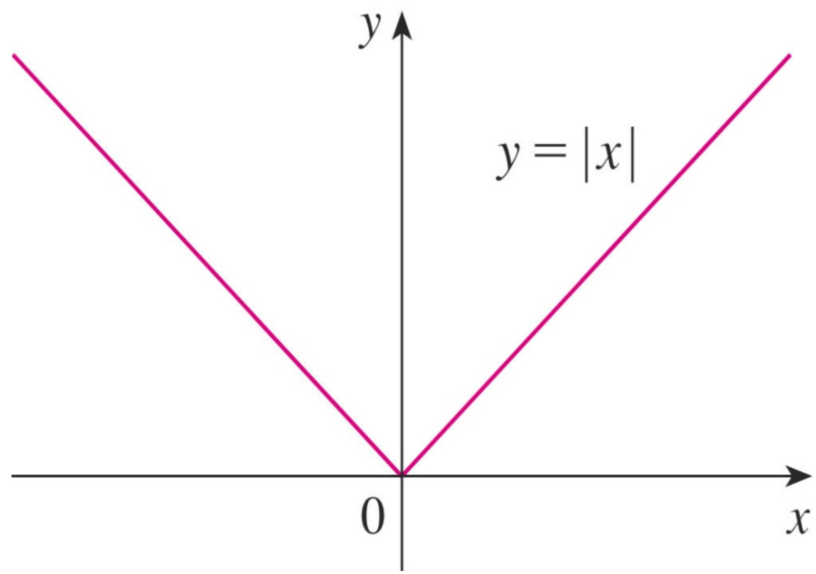
Theorem 1

$\lim_{x\to a} f(x) = L$ if and only if $\lim_{x\to a^-} f(x) = L = \lim_{x\to a^+} f(x)$.

***Example* 2.9**

Show that $\lim_{x\to 0} |x| = 0$.

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



Since $|x| = x$ for $x > 0$, we have:

$$\lim_{x\to 0^+} |x| = \lim_{x\to 0^+} x = 0$$

Since $|x| = -x$ for $x < 0$, we have:

$$\lim_{x\to 0^-} |x| = \lim_{x\to 0^-} (-x) = 0$$

Therefore, by the above Theorem 1,

$$\lim_{x\to 0} |x| = 0$$

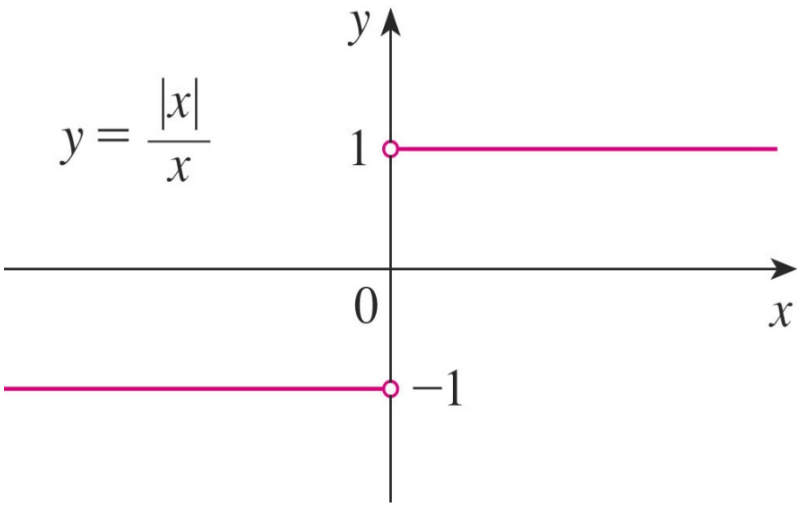
Theorem 1 (Cont')

***Example* 2.10**

Prove that $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist.

$$\lim_{x\to 0^+}\frac{|x|}{x}=\lim_{x\to 0^+}\frac{x}{x}=\lim_{x\to 0^+}1=1$$

$$\lim_{x\to 0^-}\frac{|x|}{x}=\lim_{x\to 0^-}\frac{-x}{x}=\lim_{x\to 0^-}(-1)=-1$$



Since the right- and left-hand limits are different, it follows from Theorem 1 that $\lim_{x\to 0}\dfrac{|x|}{x}$ does not exist.

Theorem 1 (Cont')

***Example* 2.11**

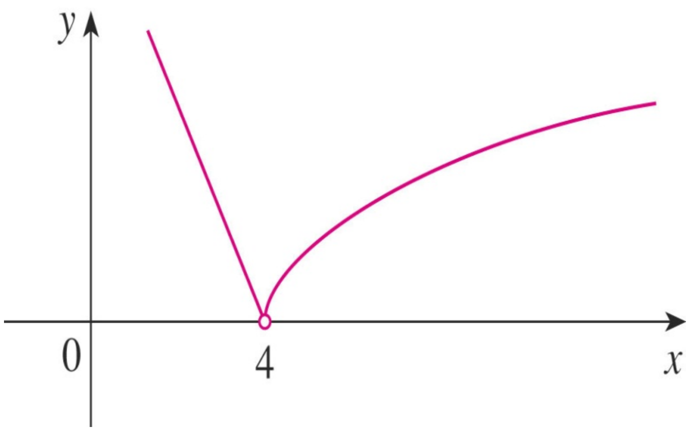
If $f(x)=\begin{cases}\sqrt{x-4} & \text{if } x>4\\ 8-2x & \text{if } x<4\end{cases}$ determine whether $\lim\limits_{x\to 4} f(x)$ exists.

- Since $f(x)=\sqrt{x-4}$ for $x > 4$,

  $\lim\limits_{x\to 4^+} f(x)=\lim\limits_{x\to 4^+}\sqrt{x-4}=\sqrt{4-4}=0$

- Since $f(x) = 8 - 2x$ for $x < 4$,

  $\lim\limits_{x\to 4^-} f(x)=\lim\limits_{x\to 4^-}(8-2x)=8-2\cdot 4=0$



- The right and left hand limits are equal,

  thus the limit exists and $\lim\limits_{x\to 4} f(x)=0$

Theorem 2

If $f(x) \leq g(x)$ when x is near a (except possibly at a) and the limits of f and g both exist as x approaches a, then

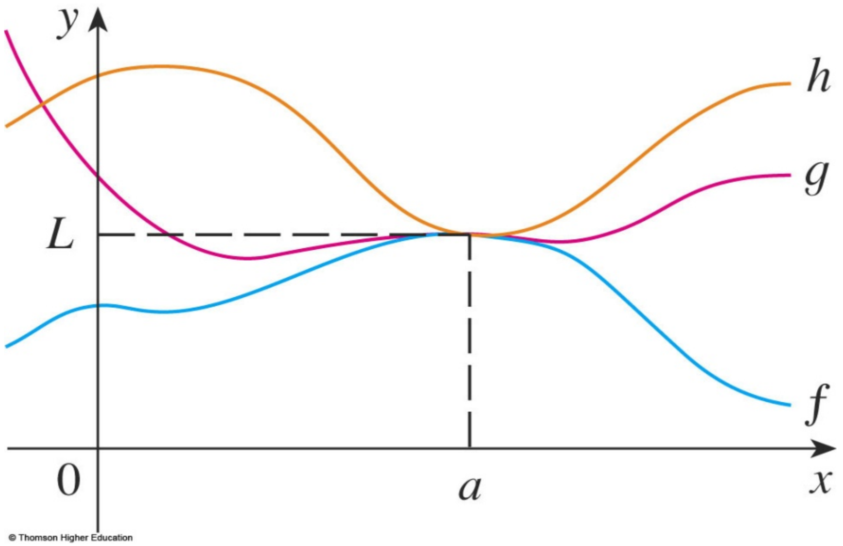
$$\lim_{x \to a} f(x) \leq \lim_{x \to a} g(x)$$

The Squeeze Theorem

If $f(x) \leq g(x) \leq h(x)$ when $x \to a$ and $\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$

then $\lim_{x\to a} g(x) = L$.

- The Squeeze Theorem is sometimes called the Sandwich Theorem or the Pinching Theorem.



The theorem is illustrated by the figure.

- It states that, if $g(x)$ is squeezed between $f(x)$ and $h(x)$ near a and if f and h have the same limit L at a, then g is forced to have the same limit L at a.

The Squeeze Theorem (Cont')

***Example* 2.12**

Find $\lim_{x\to\infty} \dfrac{\sin x}{x}$.

$$-1 \le \sin x \le 1.$$

Divide all by x: $\dfrac{-1}{x} \le \dfrac{\sin x}{x} \le \dfrac{1}{x}$.

$\lim_{x\to\infty}\left(\dfrac{-1}{x}\right) = 0$. Similarly $\lim_{x\to\infty}\left(\dfrac{1}{x}\right) = 0$.

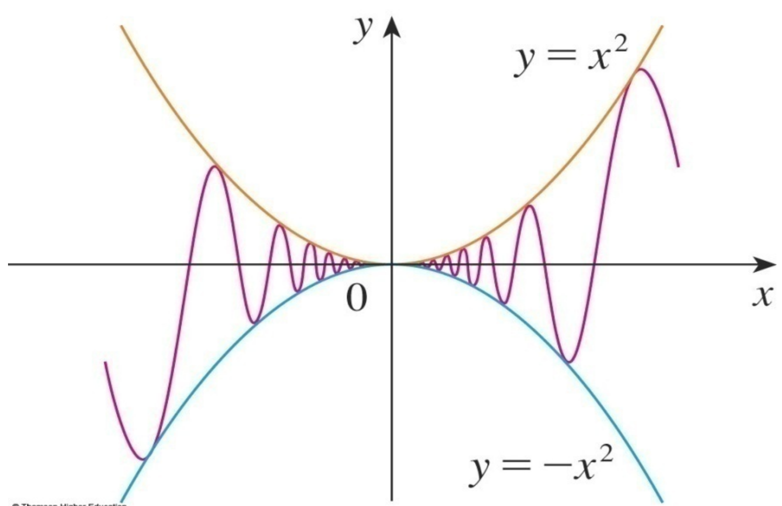
$$\therefore \lim_{x\to\infty} \frac{\sin x}{x} = 0.$$

The Squeeze Theorem (Cont')

***Example* 2.13**

Find $\lim_{x\to 0} x^2 \sin\frac{1}{x}$.

- Note that we cannot use $\lim_{x\to 0} x^2 \sin\frac{1}{x} = \lim_{x\to 0} x^2 \cdot \lim_{x\to 0} \sin\frac{1}{x}$ because $\lim_{x\to 0}\sin(1/x)$ does not exist.
- However, since $-1 \le \sin\frac{1}{x} \le 1$, we have $-x^2 \le x^2 \sin\frac{1}{x} \le x^2$.
- This is illustrated by the figure.



The Squeeze Theorem (Cont')

***Example* 2.13(Cont')**

- We know that $\lim_{x\to 0} x^2 = 0$ and $\lim_{x\to 0}(-x^2) = 0$.
- Taking $f(x) = -x^2$, $g(x) = x^2 \sin(1/x)$, and $h(x) = x^2$ in the Squeeze Theorem, we obtain:

$$\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0 .$$

Infinite Limits

Definition

Let f be a function defined on both sides of a, except possibly at a itself. Then $\lim_{x\to a} f(x) = \infty$ means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a, but not equal to a.

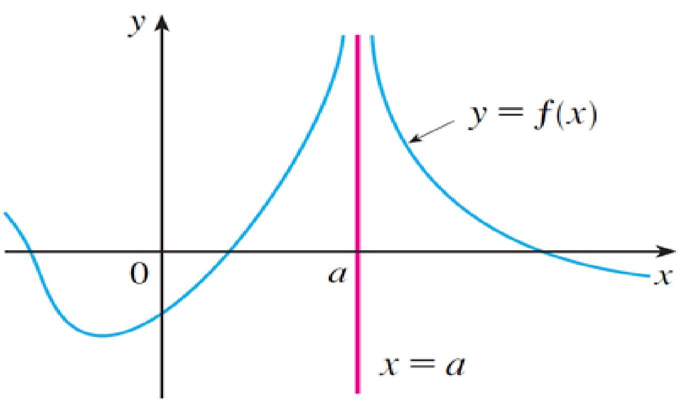


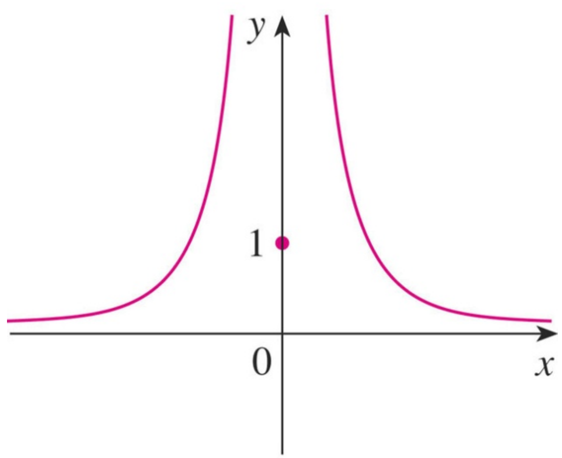
FIGURE 2
$\lim_{x\to a} f(x) = \infty$

***Example* 2.14**

Find $\lim_{x\to 0} \frac{1}{x^2}$ if it exist.

$\frac{1}{x^2} \to \infty$ as $x \to 0$.

So, $\lim_{x\to 0} \frac{1}{x^2} = \infty.$



Infinite Limits (Cont')

Definition

Let f be a function defined on both sides of a, except possibly at a itself. Then $\lim_{x \to a} f(x) = -\infty$ means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.

***Example* 2.16**

$$\lim_{x \to 0} \frac{1}{-x^2} = -\infty$$

$$-\frac{1}{x^2} \to -\infty, x \to 0$$

$$f(x) \to -\infty, x \to a$$

$$\therefore \lim_{x \to 0} \frac{1}{-x^2} = -\infty$$

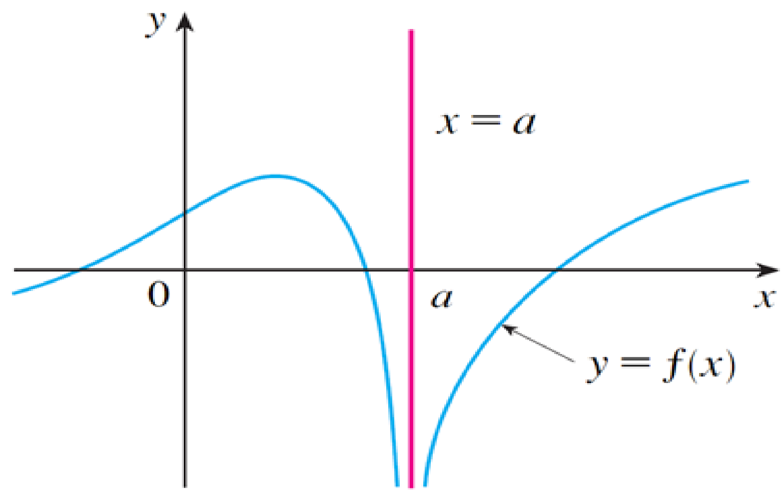


FIGURE 3

$\lim_{x \to a} f(x) = -\infty$

One-sided Infinite Limits

(1) $\lim_{x \to a^-} f(x) = \infty$ or $\lim_{x \to a^-} f(x) = -\infty$

(2) $\lim_{x \to a^+} f(x) = \infty$ or $\lim_{x \to a^+} f(x) = -\infty$

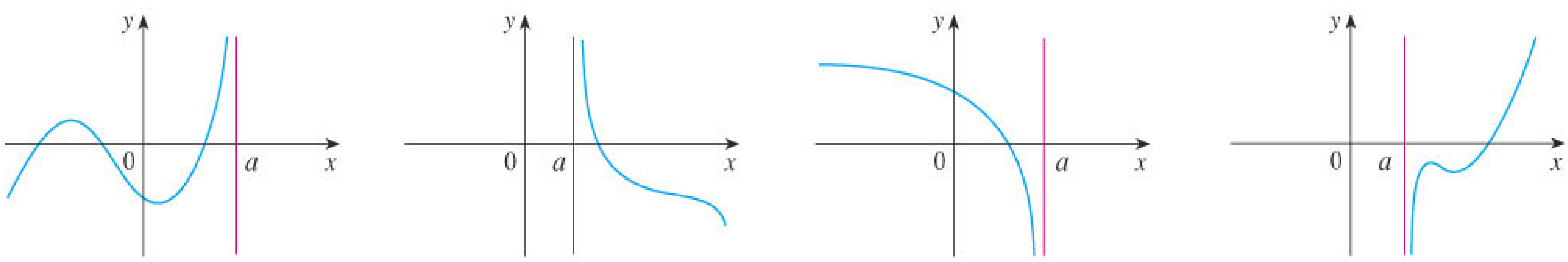
Definition

The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$\lim_{x \to a} f(x) = \infty$ $\qquad \lim_{x \to a^-} f(x) = \infty$ $\qquad \lim_{x \to a^+} f(x) = \infty$

$\lim_{x \to a} f(x) = -\infty$ $\qquad \lim_{x \to a^-} f(x) = -\infty$ $\qquad \lim_{x \to a^+} f(x) = -\infty$

- In the figure below, the line $x = a$ is a vertical asymptote in each of the four cases shown.



(a) $\lim_{x \to a^-} f(x) = \infty$ (b) $\lim_{x \to a^+} f(x) = \infty$ (c) $\lim_{x \to a^-} f(x) = -\infty$ (d) $\lim_{x \to a^+} f(x) = -\infty$

One-sided Infinite Limits (Cont')

<u>Example</u> **<u>2.17</u>**

$\lim_{x\to 0}\frac{1}{x^2}=\infty$ since $\lim_{x\to 0^+}\frac{1}{x^2}=\infty$ and $\lim_{x\to 0^-}\frac{1}{x^2}=\infty$

For this case, *y*-axis is a asymptote of $f(x)=\frac{1}{x^2}$, as $x = a$ is a vertical asymptote.

One-sided Infinite Limits (Cont')

***Example* 2.18**

Find $\lim_{x\to 3^+} \frac{2x}{x-3}$ and $\lim_{x\to 3^-} \frac{2x}{x-3}$.

- If x is close to 3 but larger than 3, then the denominator $x - 3$ is a small positive number and $2x$ is close to 6.

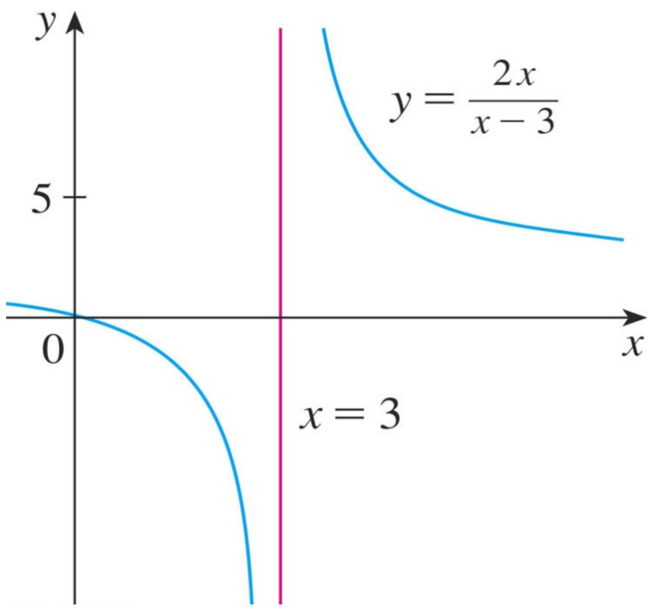
- So, the quotient $2x/(x - 3)$ is a large positive number.

- Thus, intuitively, we see that $\lim_{x\to 3^+} \frac{2x}{x-3} = \infty$.

One-sided Infinite Limits (Cont')

***Example* 2.18 (Cont')**

- Similarly, if x is close to 3 but smaller than 3, then x - 3 is a small negative number but 2x is still a positive number (close to 6).
- So, 2x/(x - 3) is a numerically large negative number.
- Thus, we see that $\lim_{x \to 3^-} \frac{2x}{x-3} = -\infty$.
- The line x – 3 is a vertical asymptote.

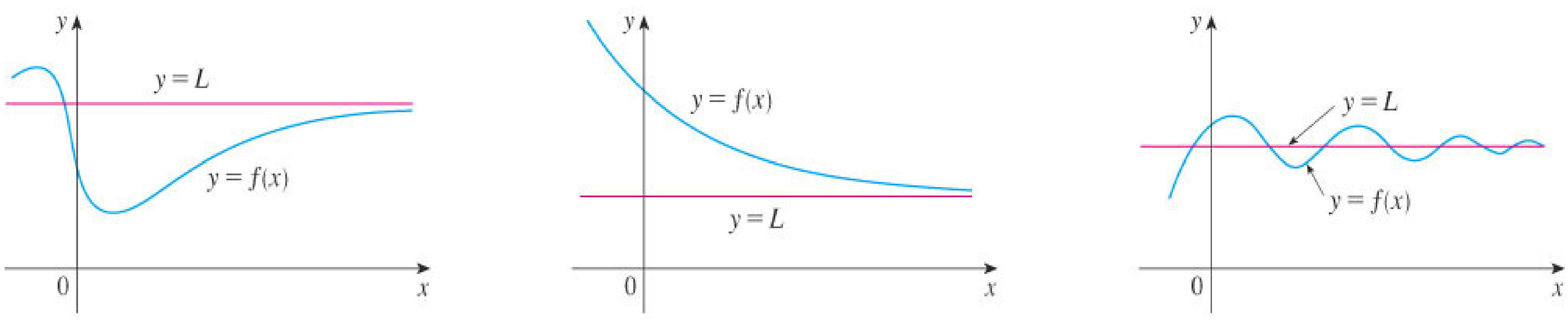


Limits when $x \to \pm\infty$ (Cont')

- If $\lim\limits_{x \to \pm\infty} f(x) = L$, then $y = L$ is the **horizontal asymptote** for the graph $y = f(x)$.

***Example* 2.19**

$f(x) \to L$ as $x \to \infty$, so $\lim\limits_{x \to \infty} f(x) = L$.

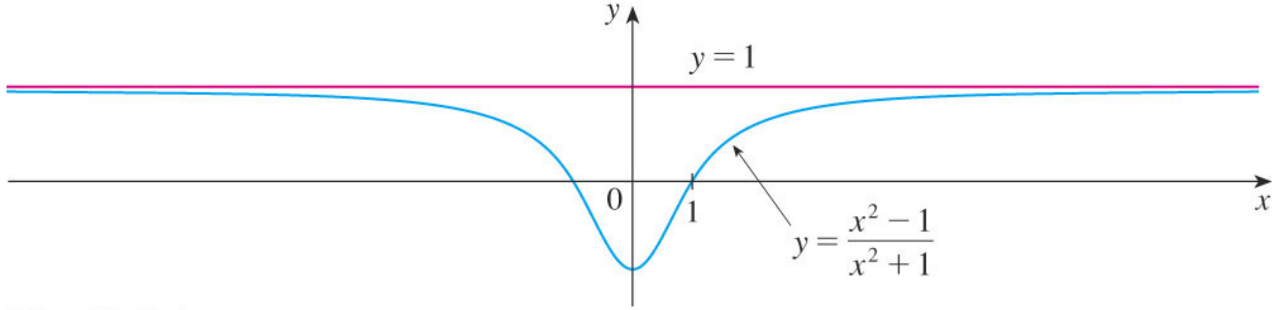


Limits when $x \to \pm\infty$ (Cont')

***Example* 2.20**

Find the horizontal asymptote for $f(x) = \dfrac{x^2 - 1}{x^2 + 1}$.

$$\lim_{x \to \infty} \frac{x^2 - 1}{x^2 + 1} = 1$$



- **The curve illustrated in the earlier figure has the line $y = 1$ as a horizontal asymptote .**
- **Note:**
- $\lim_{x \to \infty} \dfrac{1}{x^n} = 0$, for $n > 1$.

Limits when $x \to \pm\infty$ (Cont')

Example **2.21**

Evaluate $\lim\limits_{x\to\infty} \dfrac{3x^2-x-2}{5x^2+4x+1}$.

$$\lim_{x\to\infty}\frac{3x^2-x-2}{5x^2+4x+1} = \lim_{x\to\infty}\frac{\dfrac{3x^2-x-2}{x^2}}{\dfrac{5x^2+4x+1}{x^2}} = \lim_{x\to\infty}\frac{3-\dfrac{1}{x}-\dfrac{2}{x^2}}{5+\dfrac{4}{x}+\dfrac{1}{x^2}}$$

$$= \frac{\lim\limits_{x\to\infty}\left(3-\dfrac{1}{x}-\dfrac{2}{x^2}\right)}{\lim\limits_{x\to\infty}\left(5+\dfrac{4}{x}+\dfrac{1}{x^2}\right)}$$

$$= \frac{\lim\limits_{x\to\infty}3-\lim\limits_{x\to\infty}\dfrac{1}{x}-2\lim\limits_{x\to\infty}\dfrac{1}{x^2}}{\lim\limits_{x\to\infty}5+4\lim\limits_{x\to\infty}\dfrac{1}{x}+\lim\limits_{x\to\infty}\dfrac{1}{x^2}}$$

$$= \frac{3-0-0}{5+0+0}$$

$$= \frac{3}{5}$$

Try Example

Find $\lim_{x\to-\infty}\left(\dfrac{4x^2-x}{2x^3-5}\right)$.

$$\lim_{x\to-\infty}\left(\frac{4x^2-x}{2x^3-5}\right)=\lim_{x\to-\infty}\frac{\dfrac{4x^2}{x^3}-\dfrac{x}{x^3}}{\dfrac{2x^3}{x^3}-\dfrac{5}{x^3}}$$

$$=\lim_{x\to-\infty}\frac{\dfrac{4}{x}-\dfrac{1}{x^2}}{2-\dfrac{5}{x^3}}=\frac{\lim\limits_{x\to-\infty}\dfrac{4}{x}-\lim\limits_{x\to-\infty}\dfrac{1}{x^2}}{\lim\limits_{x\to-\infty}2-\lim\limits_{x\to-\infty}\dfrac{5}{x^3}}$$

$$=\frac{4\lim\limits_{x\to-\infty}\dfrac{1}{x}-\lim\limits_{x\to-\infty}\dfrac{1}{x^2}}{\lim\limits_{x\to-\infty}2-5\lim\limits_{x\to-\infty}\dfrac{1}{x^3}}=\frac{0-0}{2-0}=0$$

Try Example

Given function $f(x)=\dfrac{3x+2}{x-1}$.

(i)Find the vertical and horizontal asymptote.

(ii)Sketch the graph using information from part (i).

- Horizontal asymptotes at $x \to \pm\infty$:

$$\lim_{x\to\infty}\frac{3x+2}{x-1}=\lim_{x\to\infty}\frac{x\left(3+\dfrac{2}{x}\right)}{x\left(1-\dfrac{1}{x}\right)}=\frac{\left(3+2\lim\limits_{x\to\infty}\dfrac{1}{x}\right)}{\left(1-\lim\limits_{x\to\infty}\dfrac{1}{x}\right)}=\frac{3+0}{1-0}=3$$

$$\lim_{x\to-\infty}\frac{3x+2}{x-1}=\lim_{x\to-\infty}\frac{x\left(3+\dfrac{2}{x}\right)}{x\left(1-\dfrac{1}{x}\right)}=\frac{\left(3+2\lim\limits_{x\to-\infty}\dfrac{1}{x}\right)}{\left(1-\lim\limits_{x\to-\infty}\dfrac{1}{x}\right)}=\frac{3+0}{1-0}=3$$

We obtained that there is a horizontal asymptote $y=3$.

Try Example (Cont')

- Vertical asymptotes. A finite singular point is $x=1$ where the function is undefined.

The left-sided limit, when $x \to 1,\ x<1$ is:

$$\lim_{x\to 1^-} \frac{3x+2}{x-1} = \frac{3.1+2}{\lim\limits_{x\to 1^-}(x-1)} = \frac{5}{[-0]} = -\infty$$

The right-sided limit, when $x \to 1,\ x>1$ is:

$$\lim_{x\to 1^+} \frac{3x+2}{x-1} = \frac{5}{\lim\limits_{x\to 1^+}(x-1)} = \frac{5}{[+0]} = \infty$$

The function has two-sided vertical asymptote $x=1$.

Try Example (Cont')

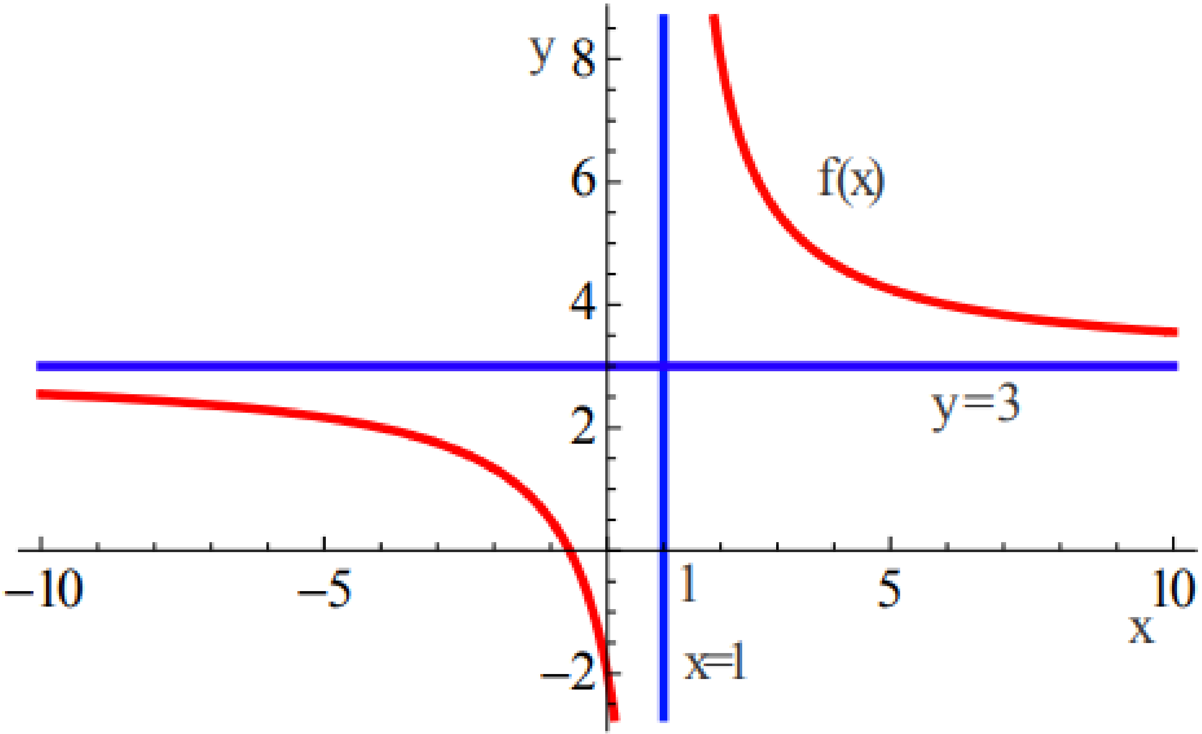


Fig. 1 Graphics of the function $f(x) = \dfrac{3x+2}{x-1}$ with its asymptotes (in blue color).

Section 2.2

Continuity

Continuity

***Definition* 1**

A function *f* is **continuous at** a **number** *a* if

$$\lim_{x \to a} f(x) = f(a).$$

Note:

- If *f* is not continuous at *a*, we say *f* is **discontinuous at** *a*, or *f* has a **discontinuity at *a***.

Continuity (cont')

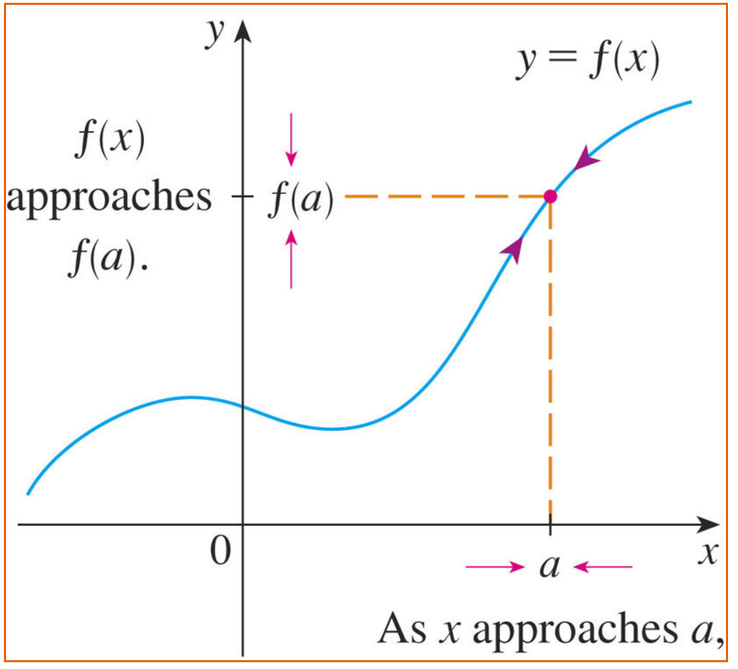
Notice that Definition 1 implicitly requires three things if f is continuous at a:

- $f(a)$ is defined—that is, a is in the domain of f
- $\lim_{x \to a} f(x)$ exists.
- $\lim_{x \to a} f(x) = f(a)$.

Continuity (cont')

The definition states that f is continuous at a if $f(x)$ approaches $f(a)$ as x approaches a.

- Thus, a continuous function f has the property that a small change in x produces only a small change in $f(x)$.
- In fact, the change in $f(x)$ can be kept as small as we please by keeping the change in x sufficiently small.



Continuity (cont')

👁 If f is defined only on one side of an endpoint of the interval, we understand 'continuous at the endpoint' to mean 'continuous from the right' or 'continuous from the left.'

Note:

- Any polynomial $y = f(x)$ is continuous for x .

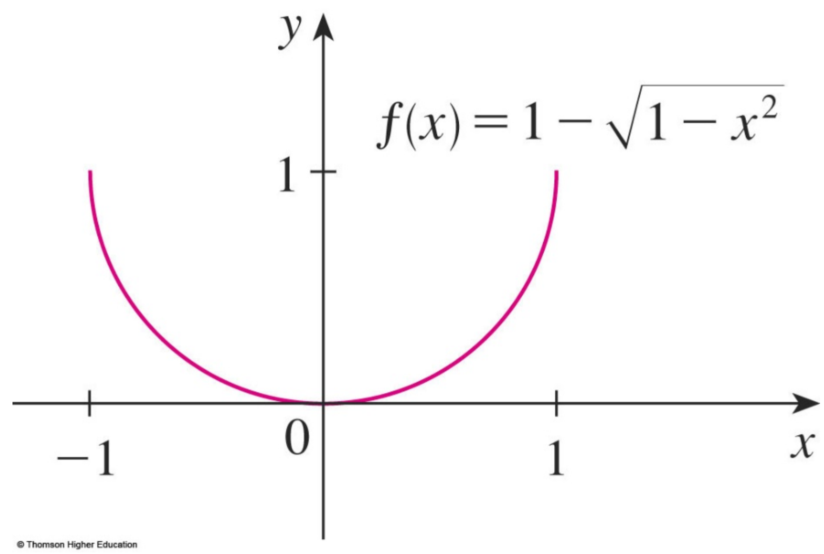
Continuity (cont')

$$\sqrt{1-x^2}$$

Continuity (cont')

The graph of *f* is sketched in the figure.

– It is the lower half of the circle $x^2 + (y-1)^2 = 1$



The Intermediate Value Theorem

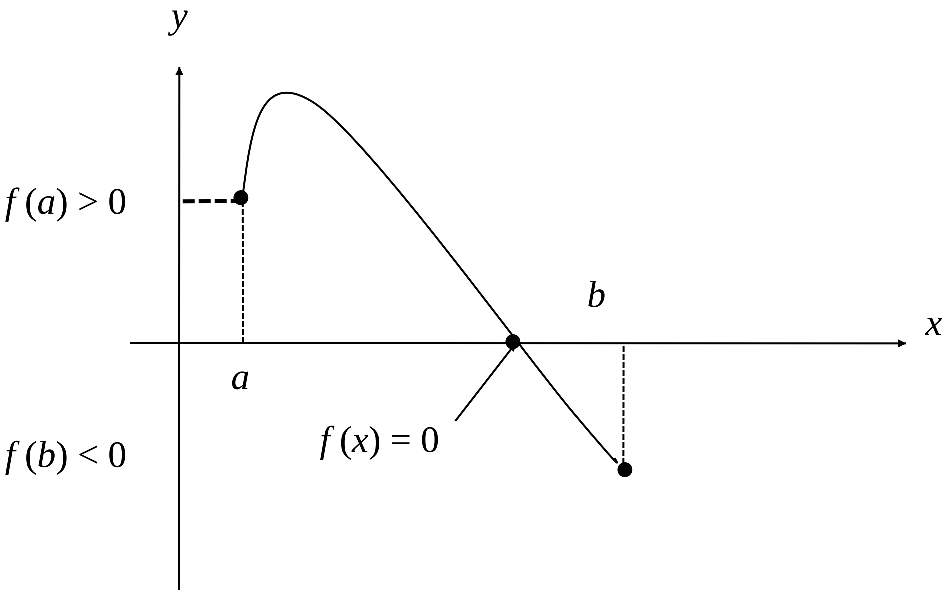
The Intermediate Value Theorem

This important property of continuous functions is proved in advanced courses.

Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

Approximating Roots Using The Intermediate Value Theorem

- If f is continuous on $[a, b]$, and if $f(a)$ and $f(b)$ are nonzero and have opposite signs, then there is at least one solution of the equation $f(x) = 0$ in the interval (a, b).



Approximating Roots Using The Intermediate Value Theorem (cont')

***Example* 2.23**

Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between 1 and 2.

- We are looking for a solution of the given equation—that is, a number c between 1 and 2 such that $f(c) = 0$.

- Therefore, we take $a = 1$, $b = 2$, and $N = 0$ in the theorem.

- We have $f(1) = 4 - 6 + 3 - 2 = -1 < 0$ and $f(2) = 32 - 24 + 6 - 2 = 12 > 0$

- Thus, $f(1) < 0 < f(2)$—that is, $N = 0$ is a number between $f(1)$ and $f(2)$.

Approximating Roots Using The Intermediate Value Theorem (cont')

***Example* 2.23(Cont')**

Now, *f* is continuous since it is a polynomial.

So, the theorem states that there is a number *c* between 1 and 2 such that $f(c) = 0$.

In other words, the equation has at least one root in the interval (1, 2).

In fact, we can locate a root more precisely by using the theorem again.

Since $f(1.2) = -0.128 < 0$ and $f(1.3) = 0.548 > 0$, a root must lie between 1.2 and 1.3.

A calculator gives, by trial and error,

$$f(1.22) = -0.007008 < 0 \text{ and } f(1.23) = 0.056068 > 0$$

So, a root lies in the interval (1.22, 1.23).

Section 2.3

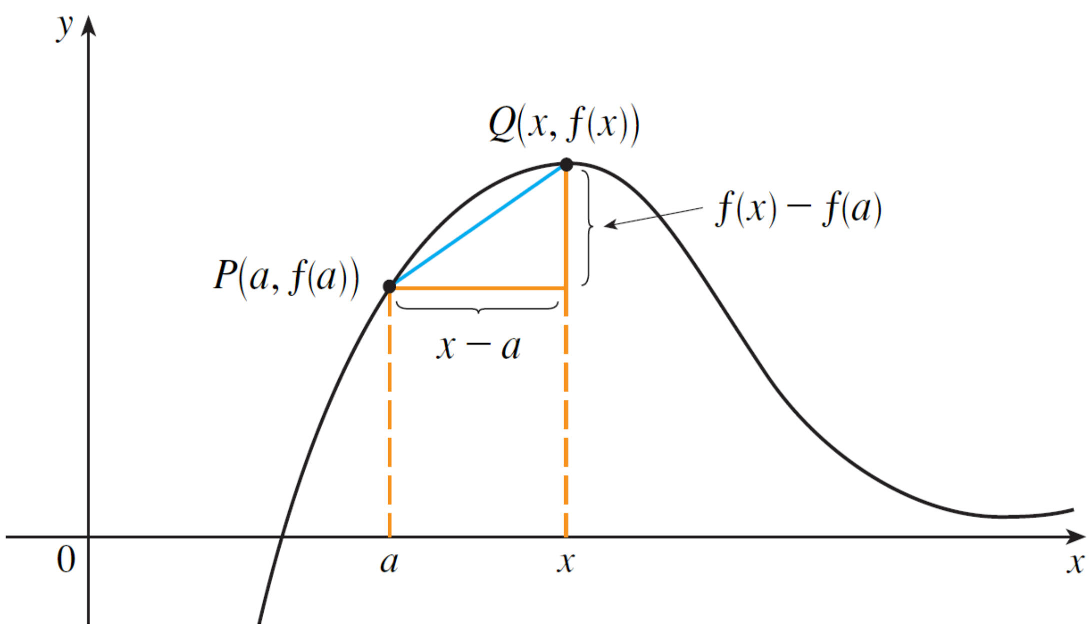
Tangents and Other Rates of Changes

Tangents

If a curve C has equation $y = f(x)$ and we want to find the tangent line to C at the point $P(a, f(a))$, then we…

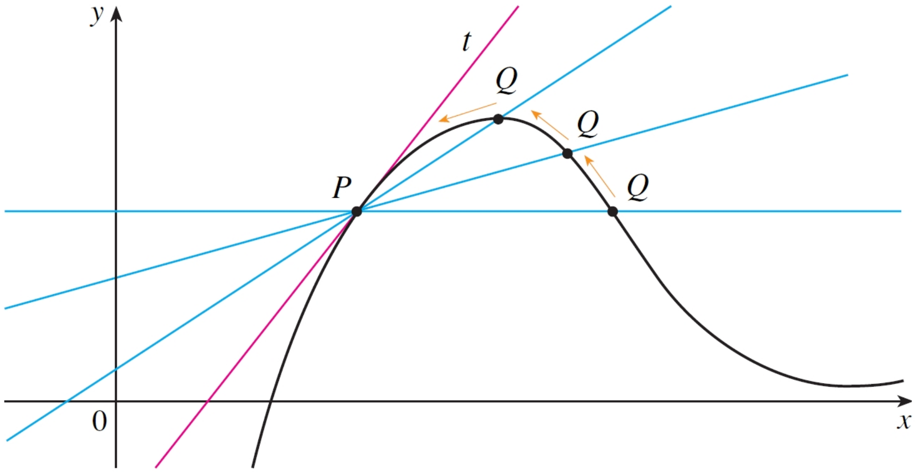
– consider a nearby point $Q(x, f(x))$, where $x \neq a$, and

– compute the slope of the secant line P, $m_{PQ} = \dfrac{f(x) - f(a)}{x - a}$



Tangents (cont')

– Then we let Q approach P along the curve C by letting x approach a.
– If m_{PQ} approaches a number m, then we define the *tangent* t to be the line
  - through P
  - with slope m.
– This saying that the tangent line is the limiting position of the secant line PQ as Q approaches P.



Tangents (cont')

– This is the content of the following definition:

1 Definition The **tangent line** to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Tangents (cont')

***Example* 2.24**

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Here $a = 1$ and $f(x) = x^2$, so the slope is

$$m = \lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \to 1} \frac{x^2 - 1}{x - 1}$$

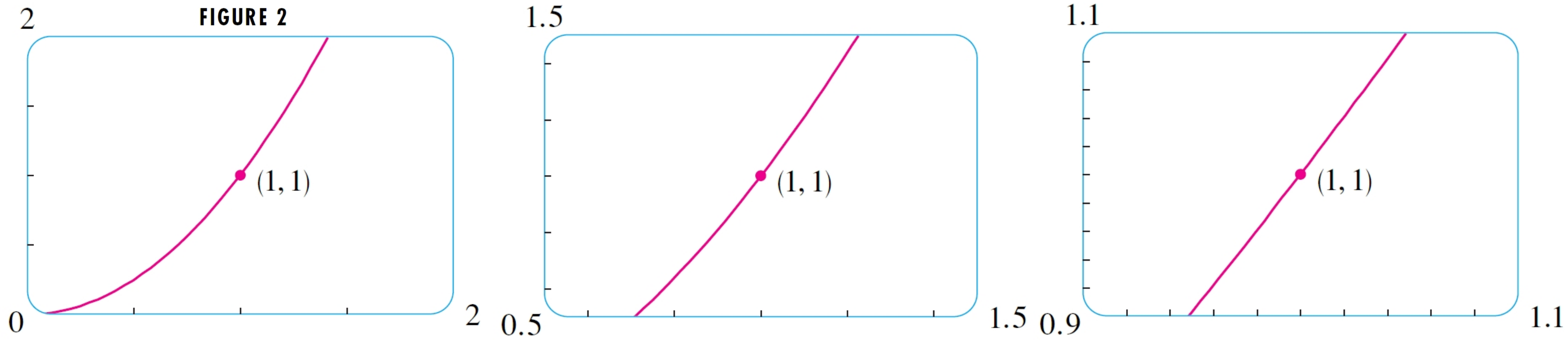
$$= \lim_{x \to 1} \frac{(x - 1)(x + 1)}{x - 1}$$

$$= \lim_{x \to 1} (x + 1) = 1 + 1 = 2$$

Tangents (cont')

- Using the point-slope form of the equation of a line, we find that an equation of the tangent line at $(1, 1)$ is
  $y - 1 = 2(x - 1)$ or $y = 2x - 1$
- The next three slides show a zoom in on the parabola toward the point $(1, 1)$.
- Note that the curve becomes almost the same as its tangent line as we zoom in:

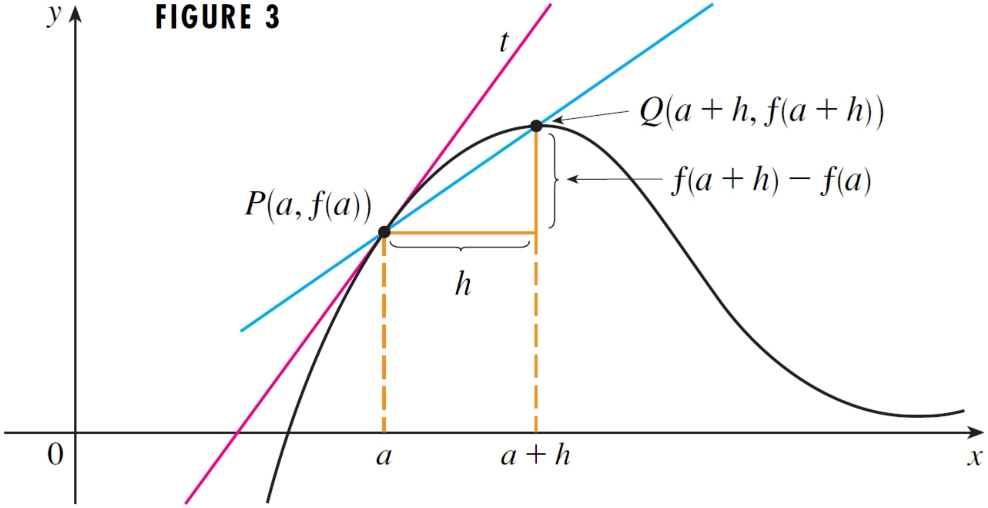
FIGURE 2



Tangents (cont')

- There is another expression for the slope of a tangent line:
  - We let $h = x - a$
  - Then $x = a + h$
  - Thus the slope of secant line PQ is $m_{PQ} = \dfrac{f(a+h) - f(a)}{h}$
- For the case $h > 0$:

FIGURE 3



- In this way the expression for the slope of the tangent line becomes:

$$m = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

Tangents (cont')

***Example* 2.25**

Find the slope of the tangent line at (1, 1) on the curve $y = x^2$.

$$m = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{(1+h)^2 - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{h^2 + 2h + 1 - 1}{h}$$

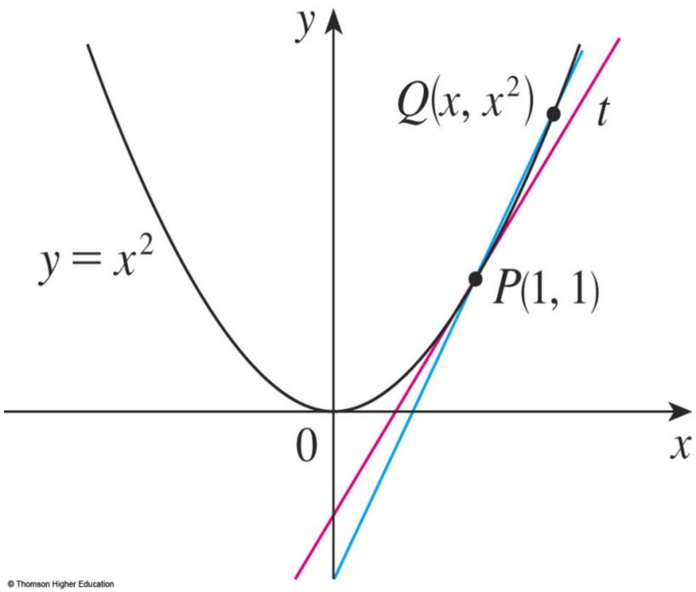
$$= \lim_{h \to 0} \frac{h^2 + 2h}{h}$$

$$= \lim_{h \to 0} \frac{h(h+2)}{h}$$

$$= \lim_{h \to 0} h + 2$$

$$= 0 + 2$$

$$= 2$$



Tangents (cont')

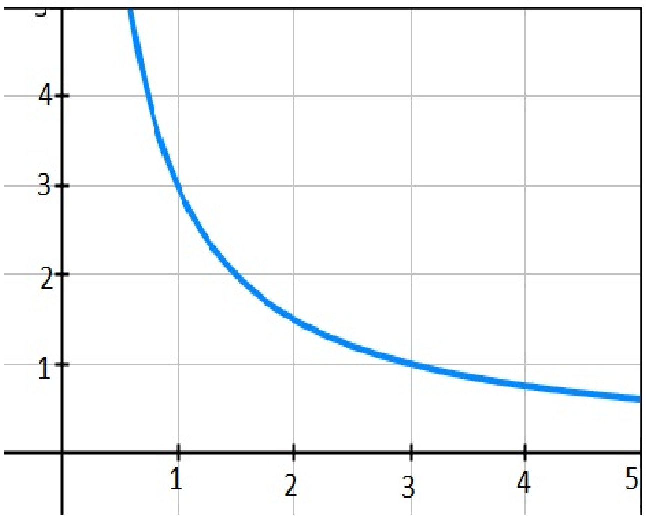
***Example* 2.26**

Find an equation of the tangent line to the hyperbola $y = \dfrac{3}{x}$ at (3,1).

$$m = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h} = \lim_{h\to 0} \frac{\dfrac{3}{3+h}-1}{h}$$

$$= \lim_{h\to 0} \frac{\dfrac{3-(3+h)}{3+h}}{h} = \lim_{h\to 0} \frac{-h}{h(3+h)}$$

$$= \lim_{h\to 0} -\frac{1}{3+h} = -\frac{1}{3}$$



Equation of the tangent at the point (3,1) is:

$$y-1 = -\frac{1}{3}(x-3)$$

$$x+3y-6=0$$

Tangents (cont')

Example **2.27**

Find the slopes of the tangent lines to the graph of the function $f(x) = \sqrt{x}$ at the points (1, 1), (4, 2), and (9, 3).

$$m = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \to 0} \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \to 0} \frac{h}{h\left(\sqrt{x+h} + \sqrt{x}\right)}$$

$$= \lim_{h \to 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}$$

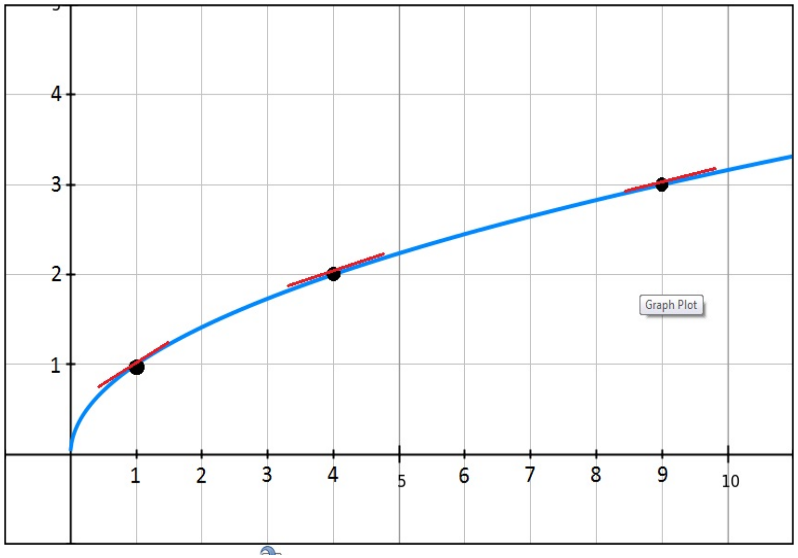
Tangents (cont')

Example 2.27(Cont')

At (1,1), $m = \dfrac{1}{2}$

At (4,2), $m = \dfrac{1}{4}$

At (9,3), $m = \dfrac{1}{6}$

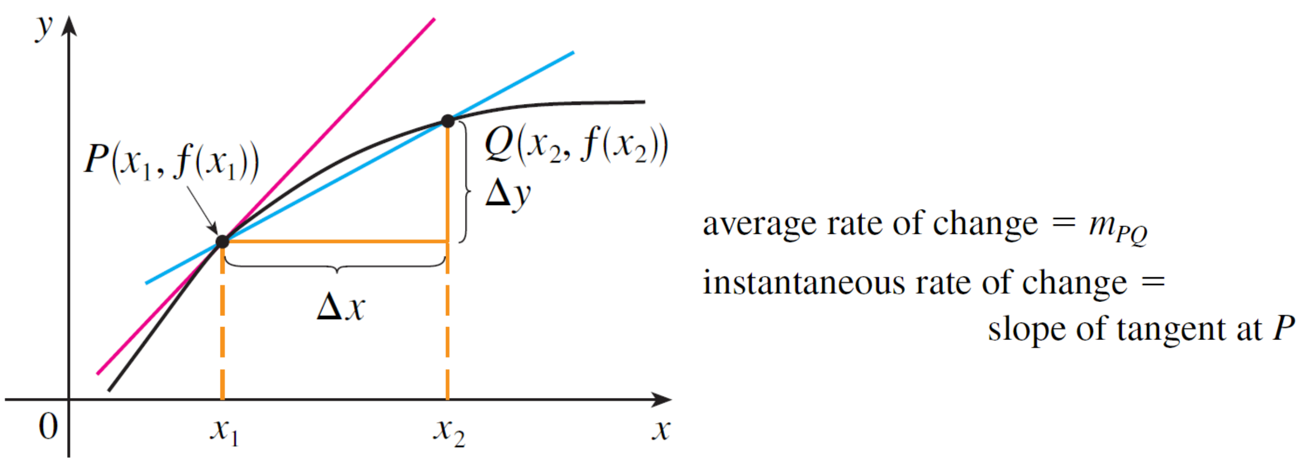


Other Rates of Change

Suppose y is a quantity that depends on another quantity x. Thus, y is a function of x and we write $y = f(x)$.

Average rate of change of y respect to x over the interval (x_1, x_2):

- If x changes from x_1 to x_2, then the change in x (increment) is $\Delta x = x_2 - x_1$
- The corresponding change in y is $\Delta y = f(x_2) - f(x_1)$.
- The difference quotient, $\dfrac{\Delta y}{\Delta x} = \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$ is called the *average rate of change of y with respect to x* over the interval $[x_1, x_2]$.
- It can be interpreted as the slope of the secant line PQ in the figure below:



Other Rates of Change (cont')

Instantaneous rate of change of *y* with respect to *x*:

- By analogy with tangent…
  - we consider the average rate over smaller and smaller intervals
  - by letting x_2 approach x_1
  - and therefore letting Δx approach 0 .
  - The limit of these average rates of change is called the (*instantaneous*) *rate of change of* y *with respect to* x at $x = x_1$.
  - The instantaneous rate of change can be interpreted as the slope of the tangent to the curve $y = f(x)$ at $P(x_1 , f(x_1))$:

$$\text{instantaneous rate of change} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\boxed{v(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}}$$

Other Rates of Change

***Example* 2.28**

Suppose that a ball is dropped from the upper observation deck of KL Tower, 450m above the ground. The distance from the tower deck for the ball, s = $4.9t^2$(m), t is the time in seconds after the fall.

(a) What is the average speed travelled from t = 2 seconds to t = 4 seconds?

(b) What is the velocity of the ball after 5 seconds?

$$v(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{4.9(x+h)^2 - 4.9x^2}{h}$$

$$= \lim_{h\to 0} \frac{4.9(2xh + h^2)}{h}$$

$$= \lim_{h\to 0} (4.9)(2)x + 4.9h$$

$$= 9.8x$$

[illegible]

(a) $v(t) = 9.8t$

$$v(t=2) = 19.6\, ms^{-1}$$

$$v(t=4) = 39.2\, ms^{-1}$$

$$\text{Average} = 29.4\ \text{ms}^{-1}$$

Section 2.4

Derivatives

Derivatives

Definition

The derivative of a function f at a number a, denoted by

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$ if this limit exists.

If $x = a + h$, $h \to 0 \Rightarrow x \to a$, then $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$.

Derivatives (cont')

Example 2.29

Find the derivative of the function $f(x) = x^2 - 8x + 9$ at $x = 1$.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$= \lim_{h \to 0} \frac{\left[(a+h)^2 - 8(a+h) + 9\right] - \left(a^2 - 8a + 9\right)}{h}$$

$$= \lim_{h \to 0} \frac{a^2 + 2ah + h^2 - 8a - 8h + 9 - a^2 + 8a - 9}{h}$$

$$= \lim_{h \to 0} \frac{2ah + h^2 - 8h}{h}$$

$$= \lim_{h \to 0} \left(2a + h - 8\right)$$

$$= 2a - 8$$

$$f'(x = 1) = -6$$

Derivatives (cont')

Interpretation of the Derivative as the Slope of a Tangent

- is the slope of the tangent line at the point $x = a$ on the curve $y = f(x)$.
- If we use the point-slope form of the equation of a line, we can write an equation of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$:

$$y - f(a) = f'(a)(x - a)$$

Derivatives (cont')

***Example* 2.30**

Find an equation of the tangent line to the parabola $y = x^2 - 8x + 9$ at the point (3,-6).

From Example 2.29, we know that the derivative of $f(x) = x^2 - 8x + 9$ at the number a is $f'(a) = 2a - 8$.

Therefore, the slope of the tangent line at (3, -6) is $f'(3) = 2(3) - 8 = -2$.

Thus, an equation of the tangent line, shown in the figure is: $y - (-6) = (-2)(x - 3)$ or $y = -2x$

Derivatives (cont')

Interpretation of the Derivative as a Rate of Change

- The derivative is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

***Example* 2.31**

The position of a particle is given by the equation of motion $s = f(t) = 1/(1 + t)$, where t is measured in seconds and s in meters. Find the velocity and the speed after 2 seconds.

$$f'(2) = \lim_{h\to 0}\frac{f(2+h)-f(2)}{h} = \lim_{h\to 0}\frac{\frac{1}{1+(2+h)}-\frac{1}{1+2}}{h} = \lim_{h\to 0}\frac{\frac{1}{3+h}-\frac{1}{3}}{h}$$

$$= \lim_{h\to 0}\frac{\frac{3-(3+h)}{3(3+h)}}{h} = \lim_{h\to 0}\frac{-h}{3(3+h)h} = -\frac{1}{9}$$

$$v(2) = -\frac{1}{9}ms^{-1}$$

$$speed = \left|-\frac{1}{9}\right| = \frac{1}{9}ms^{-1}$$

Section 2.5

The Derivative as a Function

The Derivative as a Function

- Given $y = f(x)$, derivative of y with respect to x,

$$\frac{dy}{dx} = f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- It is expressed in term of x.

***Example* 2.32**

For the function $f(x) = x^3 - x$, find $f'(x)$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{\left[(x+h)^3 - (x+h)\right] - \left[x^3 - x\right]}{h}$$

$$= \lim_{h \to 0} \frac{x^3 + 3x^2 h + 3x h^2 + h^3 - x - h - x^3 + x}{h} = \lim_{h \to 0} \frac{3x^2 h + 3x h^2 + h^3 - h}{h}$$

$$= \lim_{h \to 0} \left(3x^2 + 3xh + h^2 - 1\right) = 3x^2 - 1$$

The Derivative as a Function (Cont')

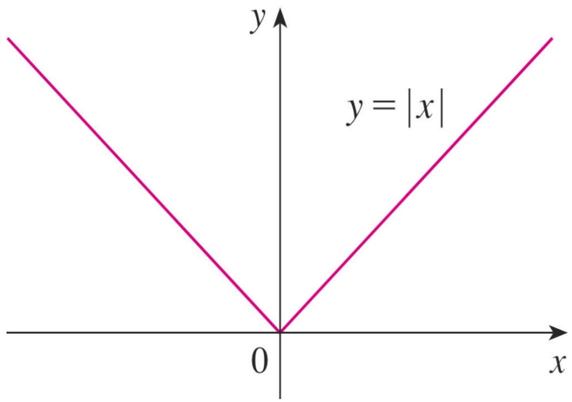
Example **2.33**

Show that $f(x) = |x|$ is not differentiable at $x = 0$.

For $x = 0$,

$$f'(0) = \lim_{h\to 0}\frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h\to 0}\frac{|0+h| - |0|}{h}$$



For right: $\lim_{h\to 0^+}\frac{|0+h|-|0|}{h} = \lim_{h\to 0^+}\frac{h}{h} = \lim_{h\to 0^+} 1 = 1$

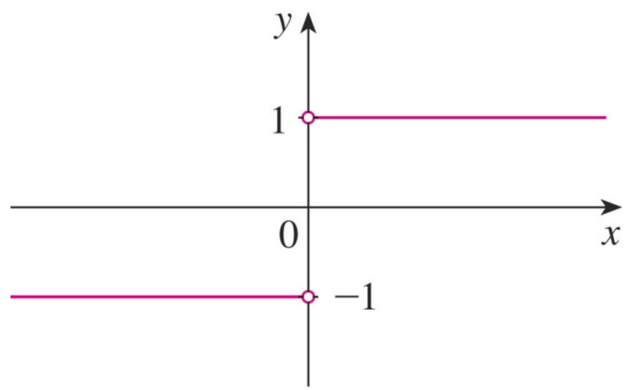
For left: $\lim_{h\to 0^-}\frac{|0+h|-|0|}{h} = \lim_{h\to 0^-}\frac{-h}{h} = \lim_{h\to 0^-}(-1) = -1$

The Derivative as a Function (Cont')

Example **2.33(Con't)**

Since these limits are different, $f'(0)$ does not exist. Thus, f is differentiable at all x except 0.

$$f'(x)=\begin{cases}1 & \text{if } x>0\\ -1 & \text{if } x<0\end{cases}$$



(b) $y = f'(x)$

Note:

- Common notations for the derivative of $y = f(x)$ are:

$$f'(x),\ y',\ \frac{dy}{dx},\ \frac{df}{dx},\ \frac{d}{dx}f(x)$$

- If f is differentiable at $x = a$, then f is continuous at $x = a$ but not vice-versa.

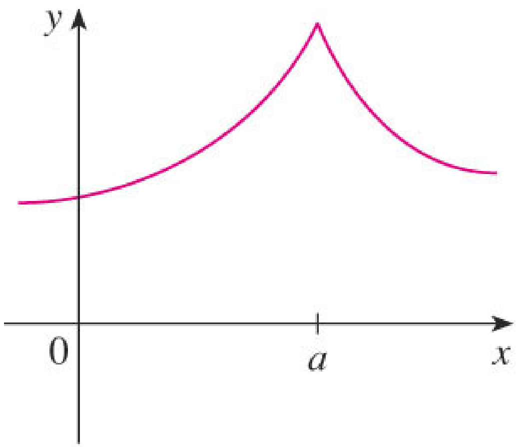
The Derivative as a Function (Cont')

Cases for $f\,'(a)$ **not exists:**

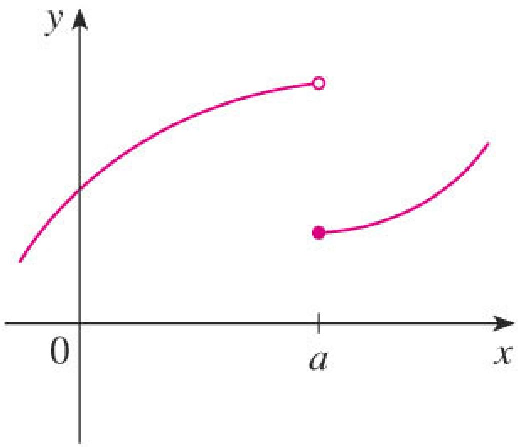
(1) The graph $y = f(x)$ has a corner at $x = a$.

(2) f is discontinuous at $x = a$.

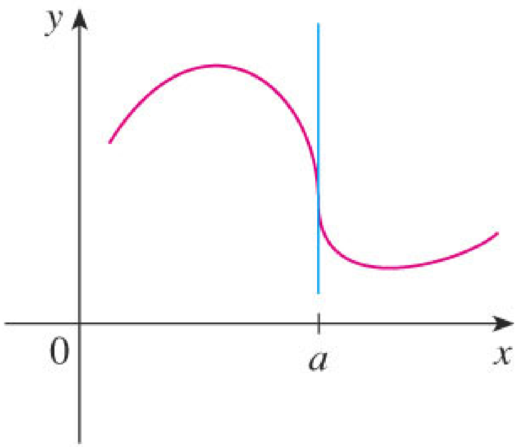
(3) The graph f has a vertical tangent line at $x = a$.



(a) A corner

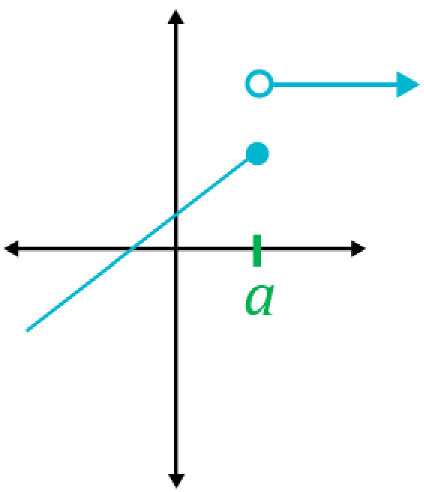


(b) A discontinuity

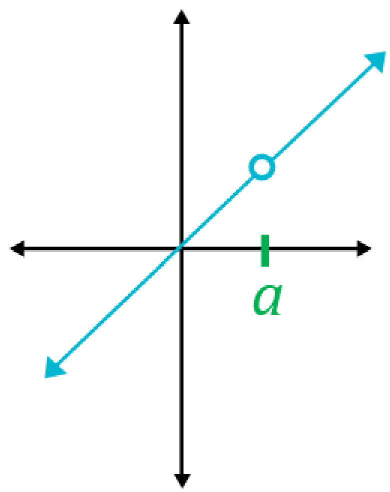


(c) A vertical tangent

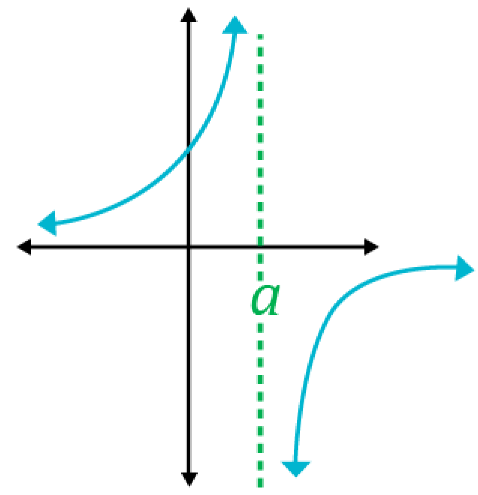
The Derivative as a Function (Cont')



Jump
Discontinuity



Removable
Discontinuity



Infinite
Discontinuity

Second Derivative

For $y = f(x)$, the second derivative of f is $\frac{d}{dx} f'(x) = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = f''(x)$.

Example 2.35

Find $f''(x)$ where $f(x) = x^3 - x$.

The first derivative is $f'(x) = 3x^2 - 1$.

So the second derivative is:

$$f''(x) = (f')'(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

$$= \lim_{h \to 0} \frac{[3(x+h)^2 - 1] - [3x^2 - 1]}{h} = \lim_{h \to 0} \frac{3h^2 + 6hx}{h}$$

$$= \lim_{h \to 0} (6x + 3h) = 6x$$

Second Derivative (Con't)

Interpretation of $y = f''(x)$

$f''(x)$ is the slope of the curve $y = f'(x)$ at the point $(x, f'(x))$.

***Example* 2.36**

Refer to Example 2.35, find $f'(2)$ and $f''(2)$.

$f'(2) = 11$

$f''(2) = 6(2) = 12$

Note:

If $f''(x) > 0$, the graph of $y = f(x)$ concaves up.

If $f''(x) < 0$, the graph of $y = f(x)$ concaves down.

Second Derivative (Con't)

n-Order Derivative for $y = f(x)$

$$y^{(n)} = f^{(n)}(x) = \frac{d^n y}{dx^n}$$

**The symbol $+\infty$ and $-\infty$ are not real numbers; do not make the mistake of manipulating these symbols using rules of algebra. For example, it is incorrect to write $+\infty - (+\infty) = 0$

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Thank you